Grade: Check Plus

Problem Set Assignment 1

Jorge Gomez

✓ Qa

(A) $P(x \geq 60) = P(60) + P(80) + P(100) = 0.15 + 0.25 + 0.05$

✓ Qb

$P(x \geq 60) = \underline{0.45}$

(B) $E(x) = 50 \quad \sigma = 26.27$

$10 \times 0.10 = 1$

$20 \times 0.15 = 3$

$40 \times 0.30 = 12$

$60 \times 0.15 = 9$

$80 \times 0.25 = 20$

$100 \times 0.05 = 5$

$E(x) = 1 + 3 + 12 + 9 + 20 + 5 = \underline{50}$

$Var(x) = E(x^2) - (E[x])^2 \qquad \sigma = \sqrt{Var(x)}$

$10^2 \times 0.10 = 100 \times 0.10 = 10$

$20^2 \times 0.15 = 400 \times 0.15 = 60$

$40^2 \times 0.30 = 1{,}600 \times 0.30 = 480$

$60^2 \times 0.15 = 3{,}600 \times 0.15 = 540$

$80^2 \times 0.25 = 6{,}400 \times 0.25 = 1{,}600$

$100^2 \times 0.05 = 10{,}000 \times 0.05 = 500$

$E[x^2] = 10 + 60 + 480 + 540 + 1{,}600 + 500 = 3{,}190$

↳ $Var(x) = 3{,}190 - 50^2$

$= 3{,}190 - 2{,}500 = 690$

↳ Standard Dev.

$\sigma = \sqrt{690} = \underline{26.27}$

(C) Mean = Expected Value

↳ Therefore we can assume that if hundreds or thousands of people withdraw money with said probabilities → the avg. withdrawal will be close to \$50 even though no single transaction will be exactly \$50.

✓ Qc

✓ Qd

(D) $\mu + \sigma = 50 \pm 26.27 \rightarrow [23.27, 76.27]$

✓ Qe

$P(23.27 \leq x \leq 76.27) = P(40) + P(60) = 0.30 + 0.15 = \underline{\underline{0.45}}$

(E) $\mu \pm 2\sigma = 50 \pm 2 \times 26.27 = 50 \pm 52.54$

$[-2.54, 102.54]$

↳ Since all possible withdrawals (10 - 100) fall within range → $P(2\sigma) = \underline{\underline{1.00}}$